When Does the Chameleon Thin-Shell Approximation Fail?

Numerical scalar-charge accuracy and force-ratio threshold maps

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Thin_Shell_Study on GitHub

1. The research gap is an error bar, not another screening demo

Known shortcut

$$Q_{\mathrm{KW}} = \min\left(3\frac{\Delta R}{R}, 1\right)$$

One number replaces a nonlinear spherical boundary-value problem.

- fast and physically transparent
- derived for a thin shell and heavy core
- usually quoted without an accuracy estimate

Question

Is the shortcut accurate enough to decide above versus below a force-ratio threshold?

1. Solve the full nonlinear field.
2. Measure its scalar charge.
3. Map approximation error.
4. Propagate error to a proxy decision boundary.

Niche: an **accuracy-to-observability bridge**, not the first thin-shell calculation.

2. A uniform sphere leaves two independent control parameters

$$V(\phi) = \frac{\Lambda^5}{\phi}, \qquad \nabla^2\phi = -\frac{\Lambda^5}{\phi^2} + \frac{\rho}{M_c}$$

$$x = \frac{r}{R}, \quad u = \frac{\phi}{\phi_\infty}, \quad \delta = \frac{\rho_c}{\rho_\infty}, \quad \mu = m_\infty R$$

Full nonlinear BVP ($n = 1$, weak-coupling limit)

$$u'' + \frac{2}{x}u' = \frac{\mu^2}{2}\left[\widetilde{\rho}(x) - u^{-2}\right]$$

$$u'(0) = 0, \qquad u(\infty) = 1$$

Diagnostics, not extra inputs

$$\epsilon = \frac{2(1-\delta^{-1/2})}{\mu^2\delta}$$

$$m_c R = \mu\delta^{3/4}$$

- $\epsilon \ll 1$: geometrically thin shell
- $m_c R \gg 1$: core can reach its minimum
- $\mu \ll 1$: source small compared with exterior range

The full exponential normalization also contains $D = \phi_\infty / M_c$; the main scan explicitly takes the same $D \to 0$ limit used by the standard formula.

3. The long-range law does not fix a finite-range source location

Long-range law

$$Q_{\mathrm{LR}} = \min(3\epsilon, 1), \quad \mu \ll 1$$

Linear shell mass plus a heuristic cap.

Two naive continuations

$$Q_{\mathrm{surf}} = Q_{\mathrm{LR}}$$

$$Q_{\mathrm{ctr}} = e^{-\mu} Q_{\mathrm{LR}}$$

Put effective charge at the surface or center.

Active-shell integral

$$g(z) = z\cosh z - \sinh z$$

$$Q_{\mathrm{AS}} = \frac{3e^{-\mu}}{\mu^3}[g(\mu) - g(\mu q)]$$

Yukawa-weights the active radial shell.

Convention warning

Surface-referenced Q carries $e^{-\mu}$; the center-referenced uniform-sphere form factor $3[\mu\cosh\mu - \sinh\mu]/\mu^3$ is greater than one.

Q_{AS} is constructed and benchmarked here, not claimed as a literature-priority formula.

4. Measuring numerical charge from the exterior tail

Charge definition

$$u = 1 - sQ_{\mathrm{num}}\frac{e^{-\mu(x-1)}}{x}, \qquad s = \frac{\mu^2\delta}{6}$$

$$Q_A(x) = \frac{x(1-u)e^{\mu(x-1)}}{s},$$

$$Q_\nabla(x) = \frac{x^2u'e^{\mu(x-1)}}{s(1+\mu x)}.$$

Agreement of two estimators is a built-in diagnostic.

- exact sharp density boundary: two collocation domains
- regular center and Yukawa Robin outer boundary
- charge-scaled variable $x(u-1) = sv$
- positive log-field representation in difficult transition points
- adaptive fourth-order collocation (SciPy `solve_bvp`)

Scan

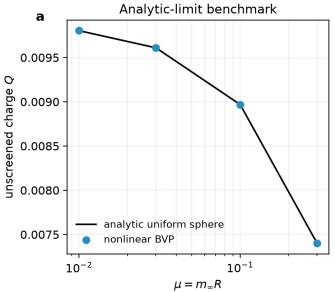
$25 \times 29 = 725$ points

$10^2 \le \delta \le 10^5$

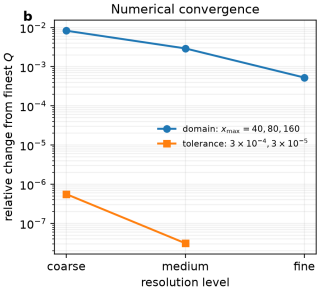
$10^{-3} \le \mu \le 0.3$, $x_{\max} = 160$

All 725 parameter points converged.

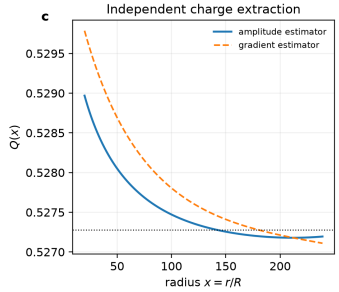
5. Verification: analytic limit, convergence, independent estimator



weak-source analytic error
2.8×10^{-5} **max**



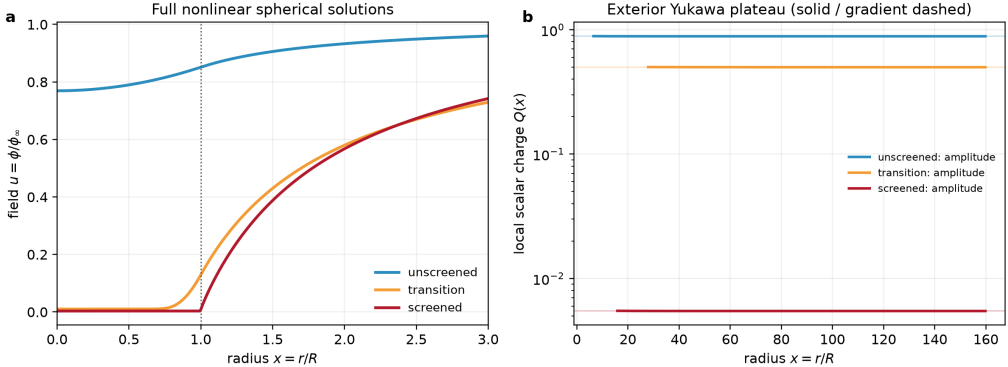
$x_{max} = 160 \rightarrow 240$ change
0.0528%



full-atlas estimator disagreement
0.0481% max

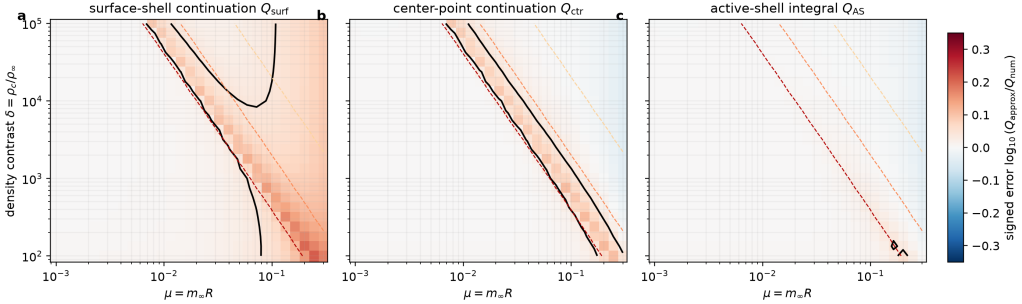
14-point stratified 160 → 240 audit: **0.0560% max** charge change, **0** flips of any 10%-accuracy classification.

6. The BVP resolves unscreened, transition, and screened regimes



$Q_{\rm num} = 0.885$ (unscreened), 0.499 (transition), and 5.47×10^{-3} (screened).
The charge plateaus are outputs of the full solution; no thin shell is imposed.

7. Headline result: finite-range continuation is not unique



Black: 10% error boundary; dashed: epsilon = 0.01, 0.1, 0.5

Red = overprediction. Black = 10% error boundary. Dashed = constant ϵ.
Surface and center continuations differ; the active-shell integral models most of the numerical trend.

8. An active-shell integral models most of the numerical trend

	Median	Maximum	$< 10\%$
Surface shell	2.78%	62.6%	68.8%
Center point	0.566%	30.1%	91.3%
Active shell	0.391%	13.1%	99.7%

Two lessons

1. 3ϵ is only first order in shell thickness.
2. Finite range requires a radial source-location model.

Checked filter for long-range 10% accuracy

$$\epsilon < 0.1, \quad m_c R > 10, \quad m_\infty R < 0.035$$

17/17 atlas + 12/12 held-out midpoints pass.

Important scope

"99.7%" means this $n = 1$, spherical, logarithmic grid. It is not a proof for all chameleon potentials or geometries.

9. A force-ratio threshold is a proxy, not an apparatus reach

Outside a spherical source

$$\alpha_5 \equiv \left|\frac{a_5}{a_N}\right| = 2\beta^2 Q_s Q_t (1+\mu x_p) e^{-\mu(x_p-1)}$$

Main map: $\beta = 1$, $Q_t = 1$, $x_p = 1.5$.

$$\alpha_\star = 10^{-2}$$

is an **illustrative dimensionless threshold**, chosen to test whether continuation ambiguity flips a proxy decision.

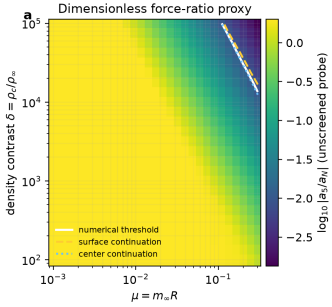
Do not conflate three statements

- α_5: physical signal ratio
- $\mathcal{R} = |a_5|/a_{\rm lim}$: reach of a past null result
- $\mathcal{D} = |\Delta a_5|/\sigma_a$: future SNR proxy

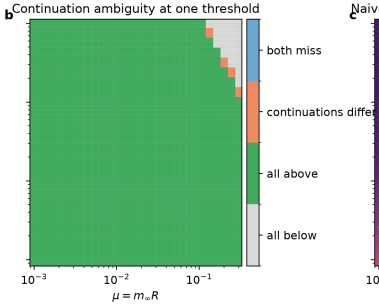
Atoms can have $Q_t \simeq 1$ while bulk probes screen, but different devices have different noise. Scalar charge alone does not rank instruments.

No source scale or detector noise is specified: this is not a sensitivity or exclusion curve.

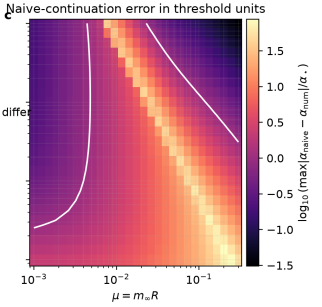
10. Continuation choice can cross the proxy threshold



continuation-sensitive points
4 of 725 at $\alpha_\star = 10^{-2}$



median contour shift
0.032 dex surface / 0.0066 dex center



convention-robust mismatch
0 points: numerical agrees with one continuation

11. What this result is—and is not

Defensible contribution

- systematic spherical transition-regime error atlas
- verified numerical scalar-charge extraction
- benchmarked active-shell finite-range ansatz
- continuation-to-threshold ambiguity map
- reproducible code, data, and tests

Not claimed

- detection of a fifth force
- a new physical (Λ, M_c) exclusion
- validity for chambers or nonspherical sources
- universal accuracy or literature priority of Q_{AS}
- revival of the basic dark-energy model

Chamber walls and three-dimensional geometry can change laboratory forces; published numerical work finds device-specific corrections at the tens-of-percent level. A 2022 levitated-sensor experiment reports ruling out the basic cosmologically viable chameleon model in its tested parameter space. Here the model is a nonlinear mechanics laboratory for approximation control.

12. Conclusion: continuation, radial structure, and one filter

1. **68.8% vs 91.3%**: surface vs center continuation within 10%.
2. **62.6%**: worst surface-continuation overprediction.
3. **99.7%**: active-shell integral within 10%.

Practical rule from this scan

A small shell is not enough. Also require a heavy core and a source small compared with the exterior Compton wavelength:

$$\epsilon < 0.1, \quad m_c R > 10, \quad m_\infty R < 0.035.$$

Research meaning

The long-range law alone cannot choose a finite-range source location. The active radial distribution controls the result.

Source, data, tests, and PDFs

`Thin_Shell_Study/`

Questions?

Selected references

[1] J. Khoury and A. Weltman, *Phys. Rev. Lett.* 93, 171104 (2004); *Phys. Rev. D* 69, 044026 (2004).

[2] T. Tamaki and S. Tsujikawa, *Phys. Rev. D* 78, 084028 (2008).

[3] C. Burrage and J. Sakstein, *Living Rev. Relativ.* 21, 1 (2018).

[4] C. Burrage, E. Copeland, and E. Hinds, *JCAP* 03, 042 (2015).

[5] B. Elder et al., *Phys. Rev. D* 94, 044051 (2016).

[6] D. Sabulsky et al., *Phys. Rev. Lett.* 123, 061102 (2019).

[7] P. Yin et al., *Nature Phys.* 18, 1181–1185 (2022).

Complete references, derivations, limitations, and reproduction instructions are in the report.